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Final Report

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AI-based Detection of Waste Littering, including cigarette butts and other litters

I. Abstract

This research investigates **AI-based detection of waste littering**, with a focus on cigarette butts and other small trash. We evaluate models on two distinct datasets: (i) a littering dataset in which most trash is extremely small, and (ii) a set of videos of people smoking where the cigarette must be detected. Prior work by other staff using state-of-the-art YOLO variants achieved **46.7% accuracy** on the littering dataset and **21%** on the cigarette dataset. We replace these with **RF-DETR**, a recent DETR-family model designed for real-time detection and obtain **~90% accuracy** on the littering dataset and **77.8% accuracy** on the cigarette dataset. These results indicate substantial headroom for further improvement and operational deployment.

II. Introduction

As summarized in the project document (Figure 1), advances in deep learning over the last decade have moved object detection beyond simple patterns matching to systems capable of sophisticated image understanding. Six major families of approaches are commonly discussed: **Convolutional Neural Networks (CNNs)**, **Transformer-based detectors**, **Vision–Language Models (VLMs)**, **hybrid designs** (e.g., RetinaMask, EfficientDet), **Sparse Coding/Dictionary Learning**, and **classic feature pipelines**. In practice, CNNs - such as the YOLO series and the R-CNN family (e.g., Mask R-CNN) - are widely deployed for their strong handling of spatial hierarchies. Transformer-based detectors like **DETR** and its variants (dynamic and deformable DETR) apply self-attention over sequences of image patches to capture global context and cast detection as set prediction, thereby removing the need for **Non-Maximum Suppression (NMS)** and simplifying post-processing.

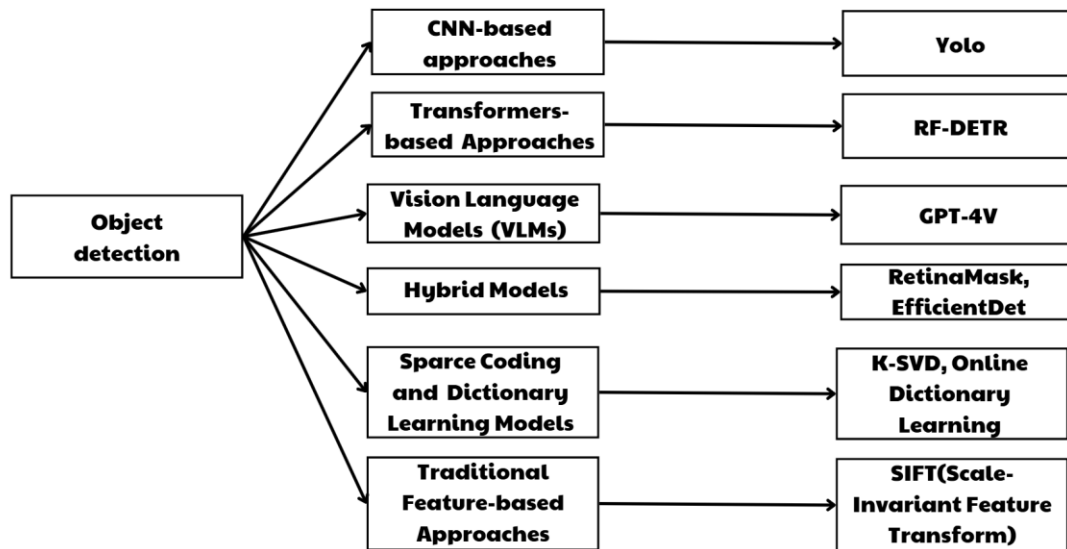


Figure 1: Classification of object detection methodologies

In reviewing related work, several comparative studies indicate that DETR-family models can outperform YOLO variants on **small-object** scenarios. For instance, a study on real-time **weed detection** reports that **RT-DETR** (a DETR-family model) outperforms YOLO for small targets, and a **green-fruit detection** study finds **RF-DETR** competitive or superior to **YOLOv12** under label ambiguity and complex orchard conditions¹. A Roboflow benchmark further notes RF-DETR as the first real-time model to surpass **60+ mAP on COCO** at lower latency, comparable to current state of the art². Motivated by these findings and our small-object focus (butts and tiny litter), we adopt **RF-DETR** to revisit prior baselines.

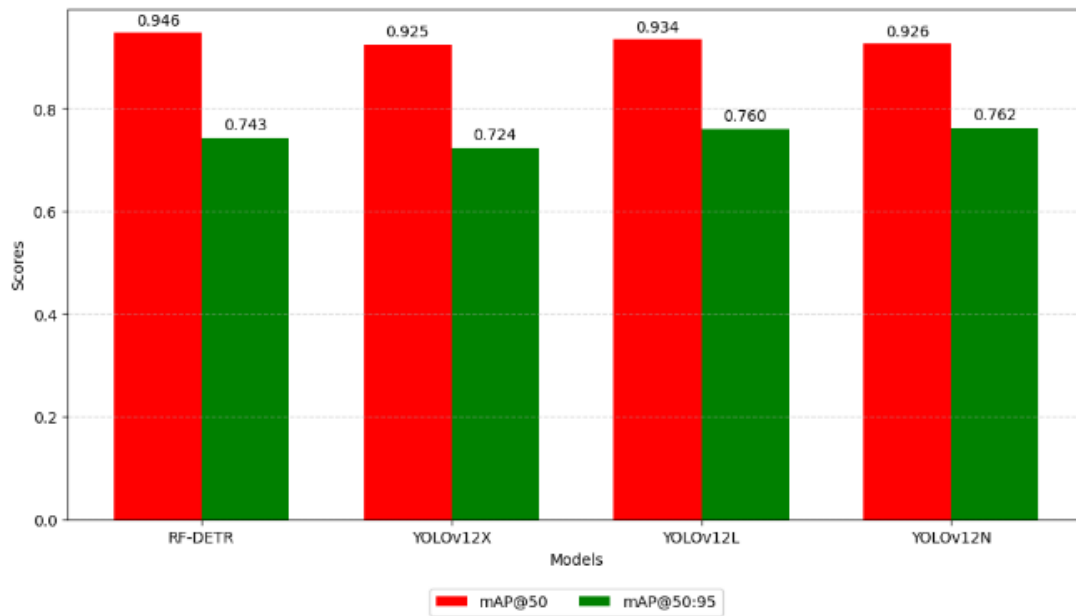
¹

Sapkota, R. C. (2025, 04 17). RF-DETR Object Detection vs YOLOv12 : A Study of Transformer-based and CNN-based Architectures for Single-Class and Multi-Class Greenfruit Detection in Complex Orchard Environments Under Label Ambiguity. doi:10.48550/arXiv.2504.13099

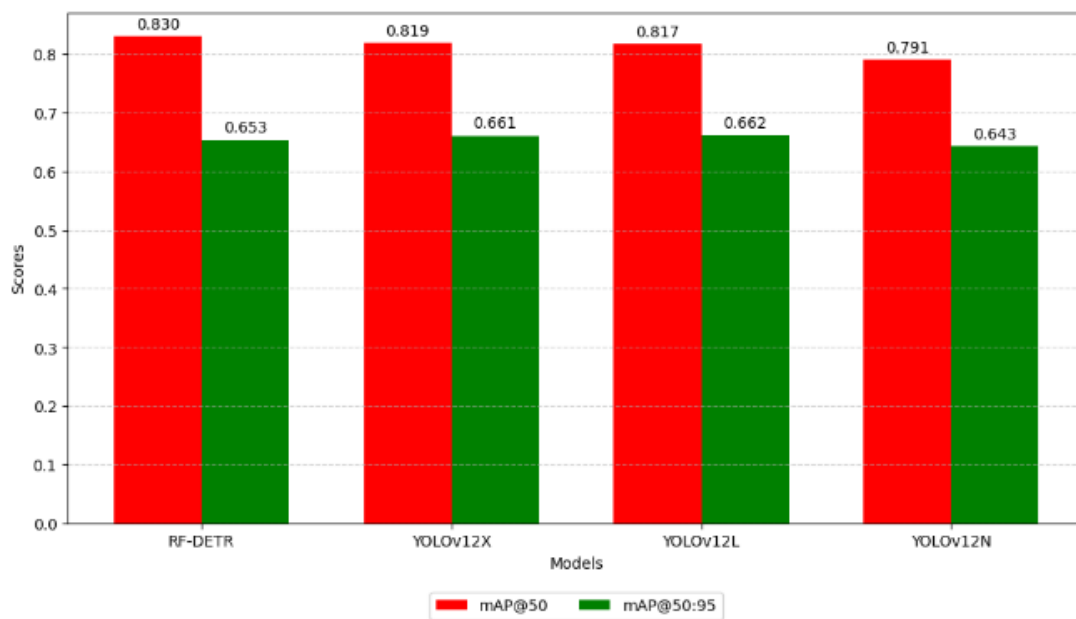
² Peter Robicheaux, J. G. (2025, 3 20). *RF-DETR: A SOTA Real-Time Object Detection Model*.

Retrieved from Roboflow Blog: <https://blog.roboflow.com/rf-detr/>:

<https://blog.roboflow.com/rf-detr/>



(a)



(b)

Figure 2: Mean Average Precision (mAP) comparison for greenfruit detection using RF-DETR and YOLOv12 object detection models: a) mAP@50 for single-class detection. b) mAP@50 and mAP@50:95 for multi-class detection.³

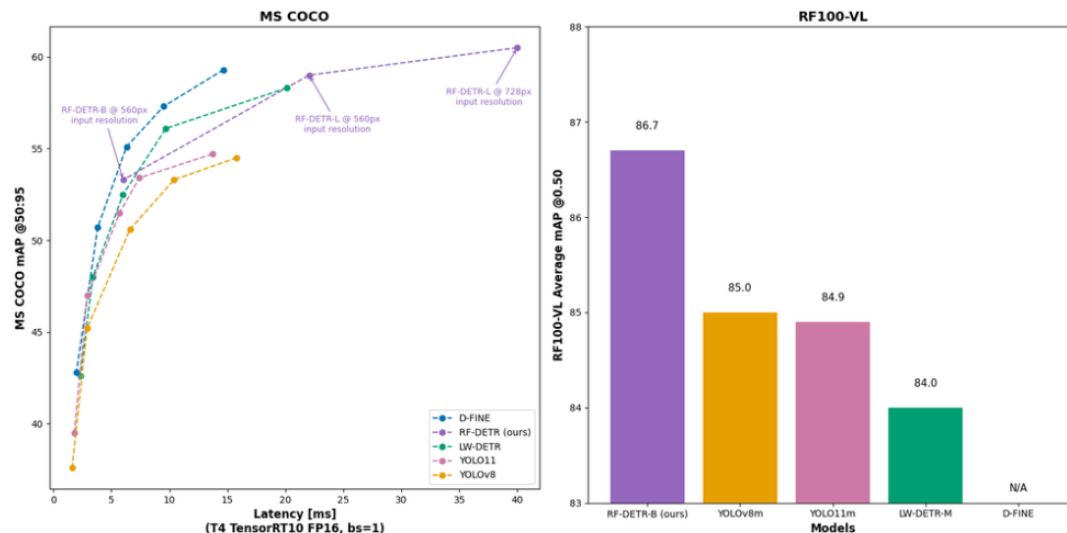


Figure 3: RF-DETR is the first realtime model to achieve 60+ mAP on COCO. Performance is comparable to current state-of-the-art at lower latencies.⁴

III. RF-DETR applied to the Cigarette dataset

A.Task Motivation

We target **cigarette-butt littering** by detecting and tracking cigarettes tossed in public areas. Cigarettes are tiny and can move quickly at discard time, making them challenging for conventional detection and tracking pipelines. Earlier interns applied **TrackNetV3** and achieved about **21% accuracy**, which failed to robustly detect cigarettes. We therefore evaluate **RF-DETR** as a stronger baseline.

B.Dataset

- **Images:** 2,500 images (training + validation)
 - **Classes:** background, cigarette
 - **Videos for test:** 9 videos
- (Example frames are shown in the original document.)

⁴ Peter Robicheaux, J. G. (2025, 3 20). *RF-DETR: A SOTA Real-Time Object Detection Model*. Retrieved from Roboflow Blog: <https://blog.roboflow.com/rf-detr/>: <https://blog.roboflow.com/rf-detr/>

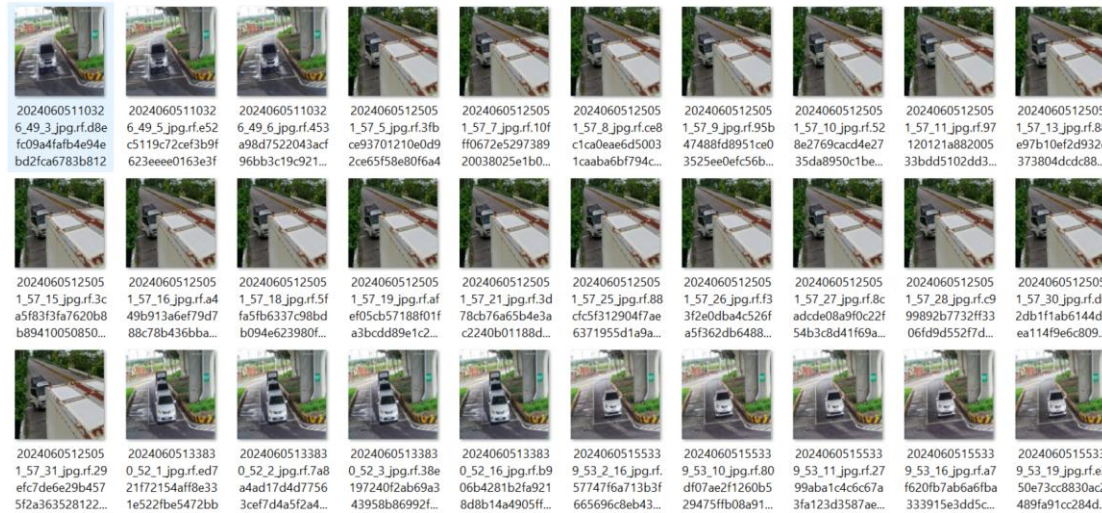


Figure 4: Examples of dataset

C.Results

With both **base** and **large** variants of RF-DETR, the best **mAP@50 = 0.40** was achieved by the **large** model. When applied to the 9 test videos, our model **successfully detected 77.8%** of smoking and cigarette-discard events - substantially outperforming the previous **TrackNetV3** result (~21%). For the two videos where the model failed, cigarettes were extremely small and partially occluded - effectively invisible to the naked eye - leaving **limited room for improvement** on the current footage.

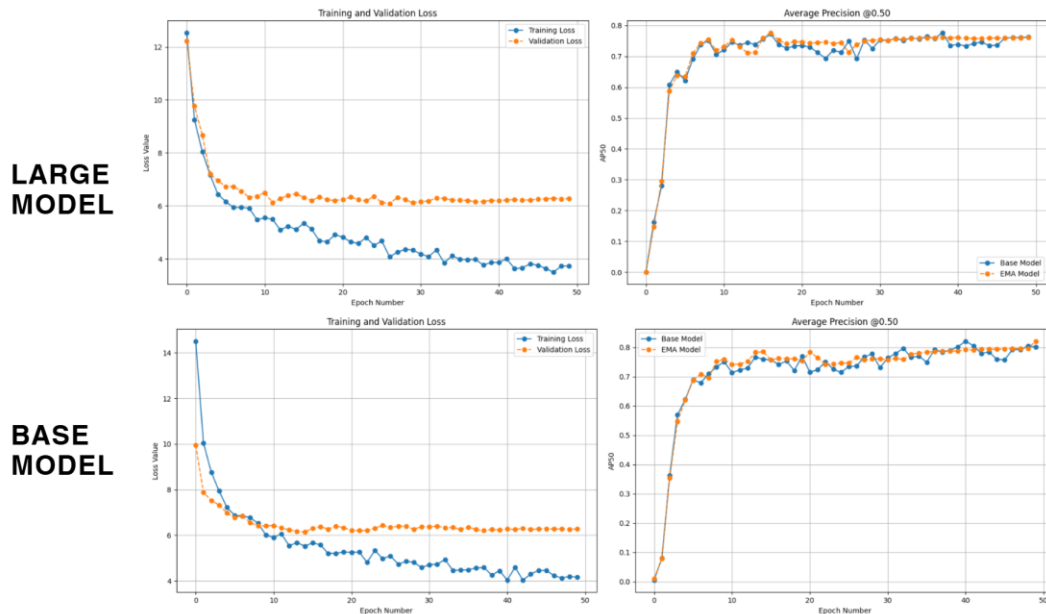


Figure 5: Model training result

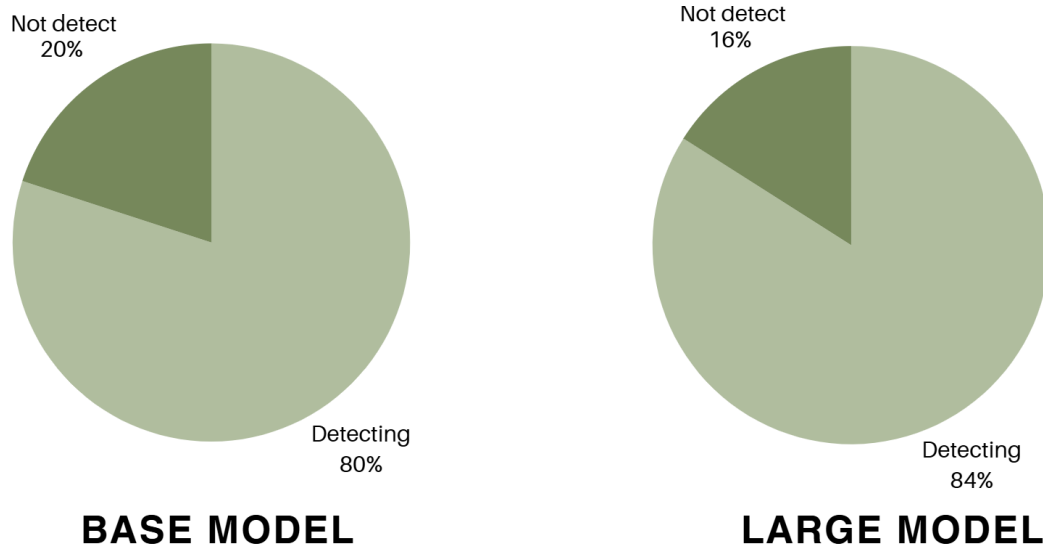


Figure 6: Model predicting result

IV. RF-DETR Applied to the Littering Dataset

A.Task & Data Preparation

The **littering dataset** contains street scenes in which pedestrians occasionally discard trash; the trash objects are **extremely small and often blurry**. An earlier team applied **YOLO + SAHI** but reached only **46.7% accuracy**. Through experimentation, we found that the original dataset included many images where objects were too small for the model to learn effectively, biasing predictions toward “small equals trash” and harming performance on larger trash.

To address this, we **combined our CCTV data** (dataset from June, July, August, September of 2024) with the **TACO** trash dataset to better balance object scales and clarity:

- **Train / Val:** 4,436 images for training and 1,169 for validation
- **Test:** 36 videos
- **Classes:** 0 and trash

This mix improves learnability while keeping the target domain aligned with littering behavior.

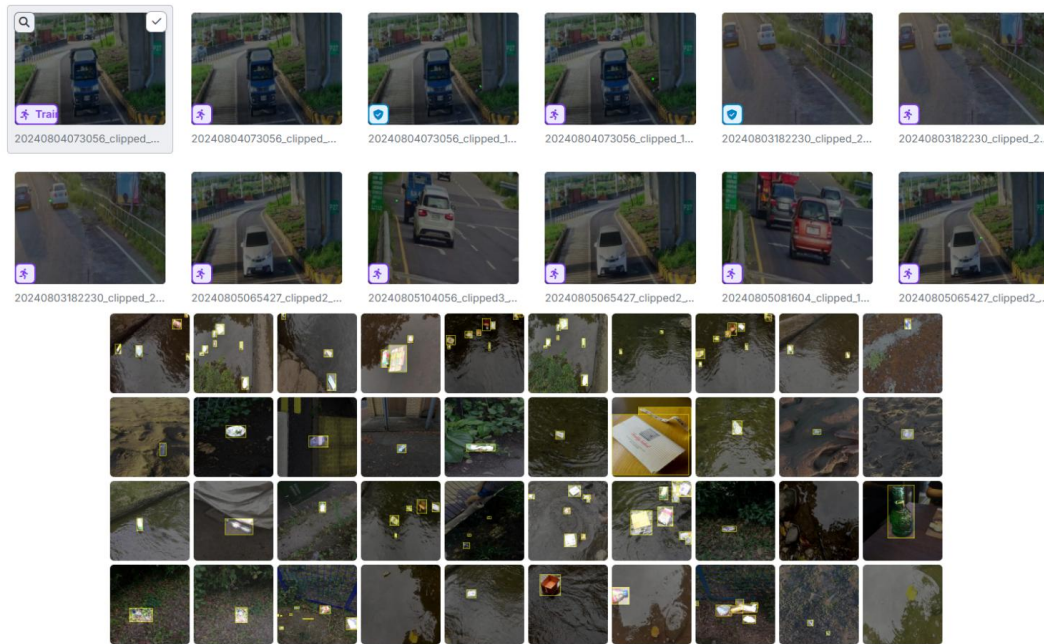


Figure 7: Examples of dataset

B.Results

On the test set, the RF-DETR model got **mAP50 of 0.675** and achieved **88.9% successful detection** of littering events, far surpassing the prior **46.7%** result.

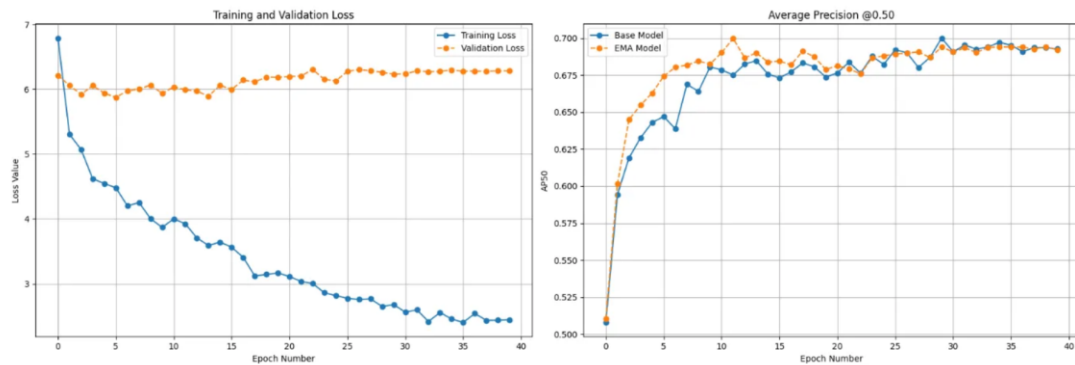


Figure 8: Training results

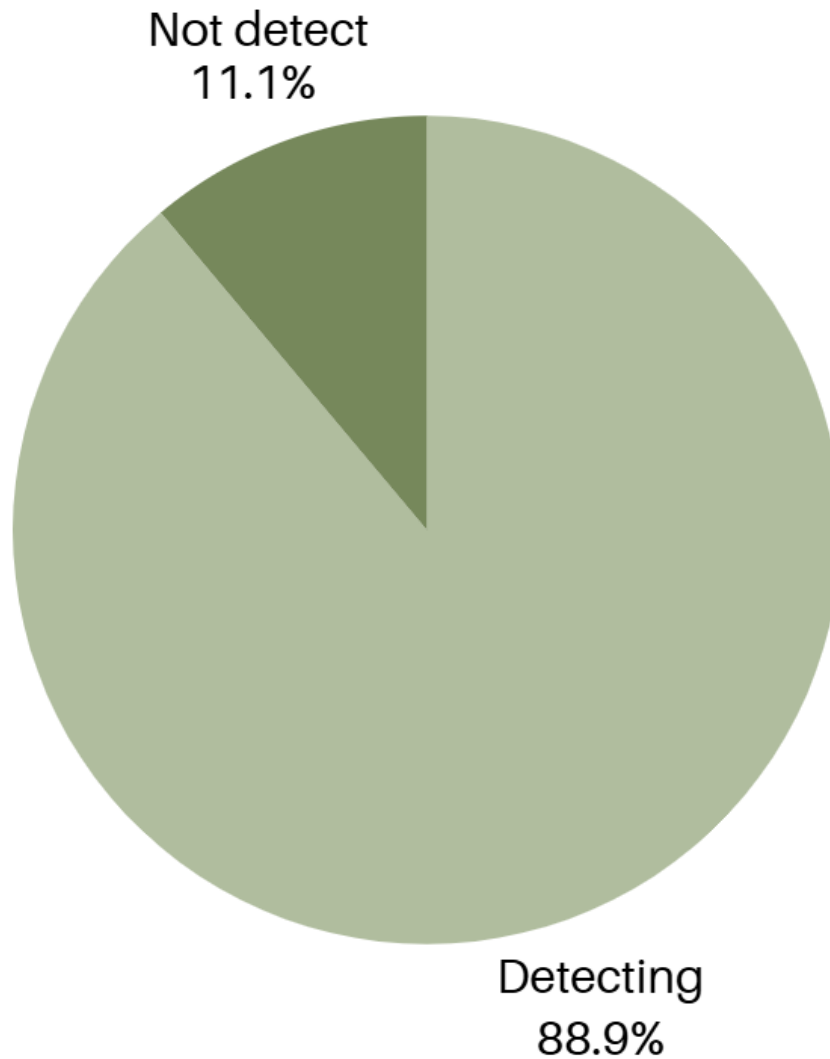


Figure 9: Predicting result

C.Limitations (Translated and Consolidated)

- **Modeling:** RF-DETR is relatively new. A promising avenue is to combine **RF-DETR with SAHI**; to our knowledge, this has not yet been reported and could further improve small-object recall.
- **Data domain gap:** TACO primarily includes **static** trash, whereas our operational target is **moving** trash (leaving a human hand). Future work should consider mixing in datasets (images or videos) that depict **clearer, larger trash during the act of discarding**, which could improve stability and overall accuracy.

V. Summary of Key Results

Table 1: Result comparision

Task / Dataset	Prior Work	My Model	Metric & Outcome
Cigarette detection (9 test videos; 2,500 train/val images; 2 classes)	TrackNetV3 (~21% accuracy)	RF-DETR (large)	mAP@50 = 0.40 (val); 77.8% successful event detection (videos)
Littering detection (36 test videos; 4,436 train / 1,169 val images; CCTV + TACO)	YOLO + SAHI (46.7% accuracy)	RF-DETR	88.9% successful event detection

VI. Discussion

Why RF-DETR helps: DETR-family models cast detection as a **set prediction** problem with global self-attention, which is especially valuable when small objects must be disambiguated from clutter. In our case, cigarettes and small litter items benefit from **global context** and improved **matching** of sparse targets. The strong uplift over YOLO-based and TrackNetV3 baselines suggests that **end-to-end Transformer** approaches are attractive for field deployment in small-object littering scenarios.

Operational implications: With ~ **90%** detection success on littering and ~**78%** on cigarette events under current conditions, RF-DETR substantially improves the feasibility of **automated monitoring**, **policy evaluation**, or **public-space analytics**. That said, **edge cases** (occlusions, minuscule scale, extreme blur) remain challenging and should guide future data collection and model design.

VII. Conclusion and Future Work

Replacing YOLO - family and TrackNetV3 baselines with **RF-DETR** yields large gains on two small-object detection tasks central to anti-littering applications. Future work

should:

1. explore **RF-DETR + SAHI** for boosted small-object recall,
2. curate **action-centric** datasets (trash leaving the hand) with **larger, clearer** instances to stabilize learning, and
3. evaluate on **broader environments** (weather, lighting, camera viewpoints) to harden models for real-world deployment.